ANALYTICAL TOOLS AND TECHNIQUES

Learning Outcome

- describe tools and techniques used in financial analysis, including their uses and limitations

The tools and techniques presented in this lesson facilitate evaluations of company data. Evaluations require comparisons. It is difficult to say that a company's financial performance was "good" or "bad" without clarifying the basis for comparison. In assessing a company's ability to generate and grow earnings and cash flow, and the risks related to those earnings and cash flows, the analyst draws comparisons to other companies at the same point in time or over the same range of time (cross-sectional analysis) and over time (trend or time-series analysis).

For example, an analyst may wish to compare the profitability of companies competing in a global industry. If the companies differ significantly in size or report their financial data in different currencies, comparing net income as reported is not useful. Ratios (which express one number in relation to another) and common-size financial statements can remove size as a factor and enable a more relevant comparison. To achieve comparability across companies reporting in different currencies, one approach is to translate all reported numbers into a common currency using average or period-end exchange rates. Alternatively, if the focus is primarily on ratios, comparability can be achieved without translating the currencies.

The analyst may also want to examine comparable performance over time. Again, the nominal currency amounts of sales or net income may not highlight significant changes. To address this challenge, horizontal financial statements (whereby quantities are stated in terms of a selected base year value) can make such changes more apparent. Another obstacle to comparison is differences in fiscal year end. To achieve comparability, one approach is to develop trailing 12 months of data. Finally, it should be noted that differences in accounting standards can limit comparability.

EXAMPLE 2

Ratio Analysis

An analyst is examining the profitability of two international companies with large shares of the global personal computer market: Acer Inc. and Lenovo Group Limited. Acer has pursued a strategy of selling its products at affordable prices. In contrast, Lenovo aims to achieve higher selling prices by stressing the high engineering quality of its personal computers for business use. Acer reports in New Taiwan dollars (TWD) and Lenovo reports in US dollars (USD). For Acer, fiscal year end is 31 December. For Lenovo, fiscal year end is 31 March; thus, FY2017 ended 31 March 2018.

The analyst collects the data shown in Exhibit 4. Use this information to answer the following questions:

Exhibit 4:

Acer versus Lenova Profitability

TWD Millions	FY2013	FY2014	FY2015	FY2016	FY2017
Revenue	360,132	329,684	263,775	232,724	237,275
Gross profit	22,550	28,942	24,884	23,212	25,361
Net income	(20,519)	1,791	604	(4,901)	2,797
Lenovo					
USD Millions	**FY2013**	**FY2014**	**FY2015**	**FY2016**	**FY2017**
Revenue	38,707	46,296	44,912	43,035	45,350
Gross profit	5,064	6,682	6,624	6,105	6,272
Net income (Loss)	817	837	(145)	530	(127)

Note: Fiscal years for Lenovo end 31 March. Thus, FY2017 represents the fiscal year ended 31 March 2018; the same applies respectively for prior years.

1. Which company is larger based on the amount of revenue, in US dollars, reported in fiscal year 2017? For FY2017, assume the relevant, average exchange rate was 30.95 TWD/USD.

Hide Solution

Solution:

Lenovo is much larger than Acer based on FY2017 revenues in US dollar terms. Lenovo's FY2017 revenues of USD45.35 billion are considerably higher than Acer's USD7.67 billion (= TWD237.275 million/30.95).

Acer: At the assumed average exchange rate of 30.95 TWD/USD, Acer's FY2017 revenues are equivalent to USD7.67 billion (= TWD237.275 million ÷ 30.95 TWD/USD).

Lenovo: Lenovo's FY2017 revenues totaled USD45.35 billion.

Note: Comparing the size of companies reporting in different currencies requires translating reported numbers into a common currency using exchange rates at some point in time. This solution converts the revenues of Acer to billions of US dollars using the average exchange rate of the fiscal period. It would be equally informative (and would yield the same conclusion) to convert the revenues of Lenovo to New Taiwan dollars.

2. Which company had the higher revenue growth from FY2016 to FY2017? FY2013 to FY2017?

Hide Solution

Solution:

The growth in Lenovo's revenue was much higher than Acer's in the most recent fiscal year and for the five-year period.

	Change in Revenue FY2016 versus FY2017 (%)	Change in Revenue FY2013 to FY2017 (%)
Acer	1.96	(34.11)
Lenovo	5.38	17.16

The table shows two growth metrics. Calculations are illustrated using the revenue data for Acer:

The change in Acer's revenue for FY2016 versus FY2017 is 1.96 percent calculated as (237,275 – 232,724) ÷ 232,724 or equivalently (237,275 ÷ 232,724) – 1. The change in Acer's revenue from FY2013 to FY2017 is a decline of 34.11 percent.

3. How do the companies compare, based on profitability?

Profitability can be assessed by comparing the amount of gross profit to revenue and the amount of net income to revenue. The following table presents these two profitability ratios—gross profit margin (gross profit divided by revenue) and net profit margin (net income divided by revenue)—for each year.

Acer	FY2013 (%)	FY2014 (%)	FY2015 (%)	FY2016 (%)	FY2017 (%)
Gross profit margin	6.26	8.78	9.43	9.97	10.69
Net profit margin	(5.70)	0.54	0.23	(2.11)	1.18

Lenovo	FY2013 (%)	FY2014 (%)	FY2015 (%)	FY2016 (%)	FY2017 (%)
Gross profit margin	13.08	14.43	14.75	14.19	13.83
Net profit margin	2.11	1.81	(0.32)	1.23	(0.28)

The net profit margins indicate that both companies' profitability is relatively low. Acer's net profit margin is lower than Lenovo's in three out of the five years. Acer's gross profit margin increased each year but remains significantly below that of Lenovo. Lenovo's gross profit margin grew from FY2013 to FY2015 and then declined in FY2016 and FY2017. Overall, Lenovo is the more profitable company, likely attributable to its larger size and commensurate economies of scale. (Lenovo has the largest share of the personal computer market relative to other personal computer companies.)

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